The Central Nervous System

Human Physiology Chapter 8

Teaching Resources

December 24, 2025

Outline

Outline

Central Nervous System Overview

- The **central nervous system (CNS)** (中枢神经系统) consists of the brain and spinal cord
- Contains enormous numbers of **association neurons** (联络神经元) and **neuroglia** (神经胶质细胞)
- Receives sensory input and directs motor output
- Performs higher functions: learning (学习), memory (记忆), emotions (情绪), logical thinking (逻辑思维)
- The brain weighs 1.5 kg and receives 15% of total blood flow

Embryonic Development

- The CNS develops from embryonic **ectoderm** (外胚层)
- A neural groove forms along the dorsal midline and deepens
- By day 20, the groove fuses to form the **neural tube** (神经管)
- The **neural crest** (神经嵴) forms between neural tube and surface ectoderm
- Neural tube → central nervous system
- Neural crest → peripheral nervous system ganglia (神经节)

Brain Region Development - Fourth Week

- By the fourth week after conception, three brain swellings appear:
  - **Forebrain** (**prosencephalon**, 前脑)
  - **Midbrain** (**mesencephalon**, 中脑)
  - **Hindbrain** (**rhombencephalon**, 后脑)
- These represent the first major differentiation of the neural tube

Brain Region Development - Fifth Week

- By the fifth week, five regions develop:
  - Forebrain divides into **telencephalon** (端脑) and **diencephalon** (间脑)
  - Midbrain remains as **mesencephalon** (中脑)
  - Hindbrain divides into **metencephalon** (后脑) and **myelencephalon** (髓脑)
- These five regions form the foundation of the adult brain structure

Brain Structure Features

- The **telencephalon** grows disproportionately in humans, forming the enormous **cerebral hemispheres** (大脑半球)
- The brain begins and remains hollow
- Brain cavities are **ventricles** (脑室), filled with **cerebrospinal fluid (CSF)** (脑脊液)
- Spinal cord cavity is the **central canal** (中央管)

Gray and White Matter

- **Gray matter** (灰质): neuron cell bodies and dendrites
  - Found in **cortex** (皮层) - surface layer
  - Found in **nuclei** (核团) - deeper aggregations
- **White matter** (白质): myelinated axon tracts
  - Underlies the cortex
  - Surrounds the nuclei
- Adult brain: 100 billion (10^11) neurons

Neurogenesis in Adults

- Adult brains contain **neural stem cells** (神经干细胞)
- **Neurogenesis** (神经发生) occurs in two locations:
  - **Subventricular zone** (室下区): newborn neurons migrate to olfactory bulbs (嗅球)
  - **Subgranular zone** (颗粒下区) in hippocampus (海马): aids hippocampus-dependent learning
- This challenges the old belief that adult brains cannot produce new neurons

Outline

Cerebrum Structure

- The **cerebrum** (大脑) is the telencephalon structure
- Largest brain portion (80% of brain mass)
- Responsible for higher mental functions
- Consists of right and left **cerebral hemispheres** (大脑半球)
- Connected by **corpus callosum** (胼胝体) - major fiber tract

Cerebral Cortex Features

- Outer **cerebral cortex** (大脑皮层): 2-4 mm of gray matter
- Characterized by **convolutions** (脑回):
  - Elevated folds: **gyri** (脑回)
  - Depressed grooves: **sulci** (脑沟)
  - Deep sulci: **fissures** (裂)
- This folding greatly increases surface area and neuron number

Cerebral Lobes

- Each hemisphere has five lobes:
  - **Frontal lobe** (额叶) - motor control, executive functions
  - **Parietal lobe** (顶叶) - sensory processing, spatial awareness
  - **Temporal lobe** (颞叶) - auditory processing, memory, language
  - **Occipital lobe** (枕叶) - visual processing
  - **Insula** (脑岛) - deep lobe, taste, visceral sensations, emotions

Lobe Functions

- Each lobe has specialized functions related to sensory processing, motor control, or higher cognitive abilities
- Functional specialization allows efficient information processing
- Damage to specific lobes produces predictable deficits

Primary Motor Cortex

- Located in **precentral gyrus** (中央前回) of frontal lobe
- **Primary motor cortex** (初级运动皮层) controls voluntary skeletal muscle movements
- Shows **somatotopic organization** (体感定位组织):
  - Body parts mapped to specific cortical areas
  - Larger areas for parts requiring fine control (hands, face)

Primary Motor Cortex

- Creates the **motor homunculus** (运动小人) - distorted body representation

Primary Somatosensory Cortex

- Located in **postcentral gyrus** (中央后回) of parietal lobe
- **Primary somatosensory cortex** (初级躯体感觉皮层) receives body sensory information
- Also shows somatotopic organization
- Larger areas for highly sensitive body parts (lips, fingertips)
- Creates the **sensory homunculus** (感觉小人)

Cortical Mapping with MRI and EEG

- Modern imaging techniques allow precise cortical mapping
- **MRI** (磁共振成像) provides structural information
- **EEG** (脑电图) provides functional activity information
- Integration of both reveals exact locations for specific functions
- Example: mapping each finger’s representation in sensory cortex

Brain Imaging: MRI

- **MRI scans** distinguish gray matter (灰质) from white matter (白质)
- Clearly show ventricles (脑室) containing cerebrospinal fluid
- Non-invasive technique using magnetic fields
- Provides detailed anatomical visualization
- Essential for diagnosing brain pathology

Brain Imaging Techniques

- Multiple techniques visualize brain structure and function:
  - **CT** (计算机断层扫描) - X-ray based
  - **MRI** (磁共振成像) - magnetic fields for structure
  - **fMRI** (功能性磁共振成像) - detects blood flow changes for activity
  - **PET** (正电子发射断层扫描) - radioactive tracers show metabolism
  - **EEG** (脑电图) - records electrical activity

Electroencephalogram (EEG)

- **EEG** records electrical brain activity through scalp electrodes
- Four normal brain wave types:
  - **Alpha waves** (α 波, 8-13 Hz) - relaxed wakefulness
  - **Beta waves** (β 波, 13-30 Hz) - alert, active thinking
  - **Theta waves** (θ 波, 4-8 Hz) - drowsiness
  - **Delta waves** (δ 波, 0.5-4 Hz) - deep sleep
- Abnormal patterns indicate disorders like epilepsy (癫痫)

Basal Nuclei Structure

- **Basal nuclei** (基底核) or **basal ganglia** (基底神经节): deep gray matter masses
- Major components:
  - **Caudate nucleus** (尾状核)
  - **Putamen** (壳核)
  - **Globus pallidus** (苍白球)
- Caudate + putamen = **corpus striatum** (纹状体)
- Connected to **substantia nigra** (黑质) in midbrain

Basal Nuclei Functions

- Control skeletal muscle movements
- Suppress unwanted movements
- Motor learning and habit formation
- Disorders:
  - **Parkinson’s disease** (帕金森病) - substantia nigra degeneration, dopamine deficiency
  - **Huntington’s disease** (亨廷顿舞蹈症) - basal nuclei neuron degeneration

Motor Circuit

- Complex interconnections between:
  - Motor cerebral cortex
  - Basal nuclei (basal ganglia, 基底神经节)
  - Thalamus (丘脑)
  - Other brain regions
- Multiple neurotransmitters:
  - **Glutamate** (谷氨酸) - excitatory
  - **Dopamine** (多巴胺) - excitatory/ modulatory
  - **GABA** (γ-氨基丁酸) - inhibitory

Cerebral Lateralization

- **Cerebral lateralization** (大脑侧化): functional differences between hemispheres
- **Left hemisphere** specializes in:
  - Language (语言)
  - Analytical thinking (分析思维)
  - Logical reasoning (逻辑推理)
  - Mathematical abilities (数学能力)
- **Right hemisphere** specializes in:
  - Spatial perception (空间感知)
  - Pattern recognition (模式识别)
  - Musical abilities (音乐能力)
  - Emotional expression (情感表达)

Split-Brain Studies

- Lateralization discovered through split-brain experiments
- Surgical severing of **corpus callosum** (胼胝体) to treat severe epilepsy
- Revealed each hemisphere’s specialized abilities
- Shows importance of hemispheric communication
- Most people are left-hemisphere dominant for language

Language Areas of the Brain

- Two main language areas (usually in left hemisphere):
  - **Broca's area** (布洛卡区, frontal lobe) - speech production (言语产生)
  - **Wernicke's area** (韦尼克区, temporal lobe) - language comprehension (语言理解)
- Connected by **arcuate fasciculus** (弓状束)
- Damage produces different types of **aphasias** (失语症):

Language Areas of the Brain

- **Broca's aphasia** (表达性失语) - difficulty producing speech
- **Wernicke's aphasia** (接受性失语) - poor comprehension, meaningless speech

Limbic System Structure

- The **limbic system** (边缘系统): interconnected structures for emotion, memory, motivation
- Major components:
  - **Hippocampus** (海马) - memory formation
  - **Amygdala** (杏仁核) - emotional processing, fear
  - **Cingulate gyrus** (扣带回) - emotion, pain
  - **Septal nuclei** (隔核) - pleasure, reward

Limbic System Structure

- **Mammillary bodies** (乳头体) - memory

Limbic System Functions

- **Emotional responses** (情绪反应)
- **Memory formation** (记忆形成), especially emotional memories
- **Motivation** (动机)
- **Olfaction** (嗅觉) - closely linked to emotion and memory
- **Autonomic regulation** (自主神经调节)
- Integration of emotion with cognition and behavior

Memory Systems

- Memory is not unitary but has multiple types:
  - **Short-term/working memory** (短期记忆/工作记忆) - brief storage, limited capacity
  - **Declarative (explicit) memory** (陈述性记忆) - consciously accessible facts and events
  - **Procedural (implicit) memory** (程序性记忆) - unconscious motor skills and habits
- Different types stored in different brain regions

Hippocampus and Memory

- **Hippocampus** (海马) is critical for forming new declarative memories
- Damage causes **anterograde amnesia** (顺行性遗忘症) - can't form new memories
- Famous case: patient H.M. demonstrated hippocampus role
- Hippocampus doesn't permanently store memories
- Helps transfer information to cortex through **memory consolidation** (记忆巩固)

Long-Term Potentiation (LTP)

- **Long-term potentiation (LTP)** (长期增强): persistent synaptic strengthening
- Cellular mechanism of learning and memory
- Involves:
  - **Glutamate** (谷氨酸) release
  - **AMPA receptors** (AMPA 受体) - Na^+ influx
  - **NMDA receptors** (NMDA 受体) - $\mathrm{Ca\hat{}(2+)}$ influx
  - Enhanced neurotransmitter release

Long-Term Potentiation (LTP)

- Structural changes including new **dendritic spines** (树突棘)

Dendritic Spines

- **Dendritic spines** (树突棘): small protrusions on dendrites
- Receive synaptic inputs
- Highly plastic structures
- Change in number and morphology during learning
- Structural basis of memory formation
- Can be seen in hippocampal neurons

Brain Areas in Emotion

- Multiple brain regions process emotions:
  - **Orbitofrontal cortex** (眶额皮层) - integrates emotion and cognition
  - **Cingulate gyrus** (扣带回) - emotional regulation
  - **Insula** (脑岛) - emotional awareness
  - **Amygdala** (杏仁核) - fear, emotional memory
- Form an emotional processing network

Emotion and Memory

- Emotions strongly influence memory formation
- **Amygdala** (杏仁核) enhances memory for emotional events
- **Emotional arousal** (情绪唤醒) leads to:
  - Enhanced attention
  - Stronger memory encoding
  - Better long-term retention
- Explains why we remember emotional events vividly

Outline

Diencephalon Overview

- **Diencephalon** (间脑) located between cerebrum and midbrain
- Three main components:
  - **Thalamus** (丘脑)
  - **Epithalamus** (上丘脑)
  - **Hypothalamus** (下丘脑)
- Each has distinct but related functions

Thalamus

- Large, paired, ovoid structure forming third ventricle walls
- Major **relay station** (中继站) for sensory information:
  - Processes and relays sensory signals to cortex (except olfaction)
  - Involved in motor function via connections with basal nuclei
  - Role in consciousness and alertness
  - Regulates sleep-wake cycles
- "Gateway to the cortex"

Epithalamus and Pineal Gland

- **Epithalamus** includes the **pineal gland** (松果体)
- Pineal gland secretes **melatonin** (褪黑素)
- Functions:
  - Regulates **circadian rhythms** (昼夜节律)
  - Controls sleep-wake cycles
  - Secretion increases during darkness
  - Decreases during light exposure

Hypothalamus Structure

- Small region below the thalamus
- Contains multiple distinct nuclei (核团):
  - **Supraoptic nucleus** (视上核)
  - **Paraventricular nucleus** (室旁核)
  - **Anterior nucleus** (前核)
  - **Posterior nucleus** (后核)
- Despite small size, has diverse critical functions

Hypothalamus Functions

- Master regulator of **homeostasis** (稳态):
  - **Autonomic control** (自主神经控制)
  - **Temperature regulation** (体温调节) - body thermostat
  - **Hunger and thirst** (饥饿和口渴)
  - **Sleep-wake cycles** (睡眠-觉醒周期)
  - **Emotional behavior** (情绪行为)
  - **Endocrine control** (内分泌控制) via pituitary gland

Hypothalamus-Pituitary Connection

- Hypothalamus controls **pituitary gland** (垂体):
  - **Posterior pituitary** (垂体后叶): stores/releases **oxytocin** (催产素) and **vasopressin/ADH** (抗利尿激素)
  - **Anterior pituitary** (垂体前叶): controlled by hypothalamic releasing/inhibiting hormones
- **Hypophyseal portal system** (垂体门脉系统) transports hormones
- **Hypothalamic-pituitary axis** (下丘脑-垂体轴) crucial for endocrine regulation

Outline

Midbrain Structure

- **Midbrain** (中脑) between diencephalon and pons
- Important structures:
  - **Cerebral peduncles** (大脑脚) - descending motor tracts
  - **Substantia nigra** (黑质) - dopaminergic neurons for motor control
  - **Red nucleus** (红核) - motor coordination
  - **Superior colliculi** (上丘) - visual reflexes
  - **Inferior colliculi** (下丘) - auditory relay
  - **Cerebral aqueduct** (中脑导水管) - connects ventricles

Dopaminergic Pathways

- Two major dopamine pathways originate in midbrain:
  - **Nigrostriatal system** (黑质纹状体系统): substantia nigra (黑质) → corpus striatum, motor control
  - **Mesolimbic system** (中脑边缘系统): midbrain → nucleus accumbens (伏隔核) and prefrontal cortex, reward and motivation
- Dysfunction relates to Parkinson's disease and schizophrenia

Hindbrain Components

- **Hindbrain** (后脑) includes:
  - **Pons** (脑桥)
  - **Medulla oblongata** (延髓)
  - **Cerebellum** (小脑)
- Each has specialized vital functions

Pons

- Contains fiber tracts connecting cerebrum, cerebellum, and spinal cord
- **Pontine respiratory centers** (脑桥呼吸中枢):
  - **Pneumotaxic center** (呼吸调整中枢)
  - **Apneustic center** (长吸中枢)
- Nuclei for cranial nerves V, VI, VII, VIII

Medulla Oblongata

- Contains ascending and descending tracts
- **Vital centers** (生命中枢):
  - **Cardiac center** (心血管中枢) - heart rate, blood vessels
  - **Respiratory rhythmicity center** (呼吸节律中枢) - breathing rhythm
  - **Vasomotor center** (血管运动中枢) - blood pressure
- **Pyramidal decussation** (锥体交叉) - corticospinal fibers cross
- Nuclei for cranial nerves IX, X, XI, XII

Respiratory Control Centers

- Nuclei in pons and medulla control breathing
- Multiple centers coordinate respiratory rhythm
- Essential for maintaining proper gas exchange
- Integration with other systems for homeostasis
- Damage can be fatal

Reticular Activating System

- **Reticular formation** (网状结构): neuron network through medulla, pons, midbrain
- **Reticular activating system (RAS)** (网状激活系统) projects to thalamus and cortex
- Functions:
  - Maintains **consciousness** (意识) and **alertness** (警觉)
  - Regulates **sleep-wake cycles** (睡眠-觉醒周期)
  - Filters sensory information

Reticular Activating System

- Controls **attention** (注意力) and **arousal** (觉醒)
- Damage causes **coma** (昏迷)

Cerebellum

- Second largest brain region
- Located dorsal to pons and medulla
- Highly folded surface
- Connected to brain stem by three pairs of **cerebellar peduncles** (小脑脚)
- Essential for coordinated movement

Cerebellum Functions

- **Motor coordination** (运动协调)
- **Balance** (平衡) and **posture** (姿势)
- **Motor learning** (运动学习) - fine-tuning through practice
- **Timing** (时间控制) of movements
- Cognitive functions - planning, language
- Damage causes **ataxia** (共济失调), **intention tremor** (意向性震颤), **dysmetria** (测量障碍)

Outline

Overview of Spinal Tracts

- Spinal cord contains:
  - **Ascending tracts** (上行传导束) - sensory information to brain
  - **Descending tracts** (下行传导束) - motor commands from brain
- Most tracts exhibit **decussation** (交叉) - crossing to opposite side
- Explains **contralateral control** (对侧控制) - each hemisphere controls opposite body side

Ascending Tracts Overview

- Carry sensory information from periphery to brain
- Most involve three-neuron pathways:
  - **First-order neuron** (一级神经元): receptor → spinal cord/brain stem
  - **Second-order neuron** (二级神经元): spinal cord/brain stem → thalamus (crosses midline)
  - **Third-order neuron** (三级神经元): thalamus → cerebral cortex

Principal Ascending Tracts

- Major ascending pathways:
  - **Fasciculus gracilis/cuneatus** (薄束/楔束): discriminative touch, proprioception (本体感觉)
  - **Lateral spinothalamic tract** (外侧脊髓丘脑束): pain, temperature
  - **Anterior spinothalamic tract** (前脊髓丘脑束): crude touch
  - **Spinocerebellar tracts** (脊髓小脑束): unconscious proprioception to cerebellum

Ascending Tracts Pathways

- **Medial lemniscal tract** (内侧丘系束): carries touch, pressure, proprioception
- **Lateral spinothalamic tract** (外侧脊髓丘脑束): carries pain and temperature
- Both pathways decussate (交叉) at different levels
- Third-order neurons deliver information to cerebral cortex
- Allows precise sensory localization

Descending Tracts Overview

- Carry motor commands from brain to spinal motor neurons
- Two main systems:
  - **Pyramidal tracts** (锥体束) or **corticospinal tracts** (皮质脊髓束)
  - **Extrapyramidal tracts** (锥体外系)
- Work together for coordinated voluntary and involuntary movements

Descending Motor Tracts

- **Pyramidal tracts**: direct voluntary control
- **Extrapyramidal tracts**: posture, balance, involuntary movements
- Multiple pathways from different brain regions
- Converge on spinal motor neurons
- Integration allows complex coordinated movements

Pyramidal (Corticospinal) Tracts

- Originate from **primary motor cortex** (初级运动皮层) in precentral gyrus
- Two divisions:
  - **Lateral corticospinal tract** (外侧皮质脊髓束): 85-90% cross at pyramidal decussation, control limbs
  - **Anterior corticospinal tract** (前皮质脊髓束): cross at spinal level, control axial muscles
- Provide direct voluntary control of skeletal muscles

Pyramidal (Corticospinal) Tracts

- Damage causes paralysis (瘫痪) and loss of fine motor control

Extrapyramidal Tracts

- Originate from brain stem nuclei
- Major tracts:
  - **Rubrospinal tract** (红核脊髓束): from red nucleus
  - **Tectospinal tract** (顶盖脊髓束): from superior colliculus
  - **Vestibulospinal tract** (前庭脊髓束): from vestibular nuclei
  - **Reticulospinal tract** (网状脊髓束): from reticular formation
- Control posture, balance, involuntary movements

Integration of Motor Systems

- **Pyramidal tracts** (shown in pink) provide voluntary motor control
- **Extrapyramidal tracts** from brain stem control posture and involuntary movements
- Both systems converge on lower motor neurons in spinal cord
- Integration allows smooth, coordinated movements
- Balance between systems essential for normal motor function

Outline

Cranial Nerves Overview

- 12 pairs of **cranial nerves** (脑神经) arise from brain
- Numbered I through XII (Roman numerals)
- Can be sensory, motor, or mixed (both)
- Serve head, neck, and some thoracic/abdominal organs
- Essential for special senses, facial movement, and autonomic functions

Summary of Cranial Nerves

- I **Olfactory** (嗅神经) - smell
- II **Optic** (视神经) - vision
- III **Oculomotor** (动眼神经) - eye movements
- IV **Trochlear** (滑车神经) - eye movement
- V **Trigeminal** (三叉神经) - facial sensation, chewing
- VI **Abducens** (外展神经) - eye movement

More Cranial Nerves

- VII **Facial** (面神经) - facial expression, taste
- VIII **Vestibulocochlear** (前庭蜗神经) - hearing, balance
- IX **Glossopharyngeal** (舌咽神经) - taste, swallowing
- X **Vagus** (迷走神经) - parasympathetic to organs
- XI **Accessory** (副神经) - neck/shoulder movements
- XII **Hypoglossal** (舌下神经) - tongue movements

Spinal Nerves Organization

- 31 pairs of **spinal nerves** (脊神经), all mixed
- Named by vertebral level:
  - 8 **cervical** (颈神经) C1-C8
  - 12 **thoracic** (胸神经) T1-T12
  - 5 **lumbar** (腰神经) L1-L5
  - 5 **sacral** (骶神经) S1-S5
  - 1 **coccygeal** (尾神经)
- Each formed by union of dorsal (sensory) and ventral (motor) roots

Spinal Nerve Distribution

- Spinal nerves interconnect at **nerve plexuses** (神经丛):
  - **Cervical plexus** (颈丛) C1-C4: neck, diaphragm
  - **Brachial plexus** (臂丛) C5-T1: upper limb
  - **Lumbar plexus** (腰丛) L1-L4: anterior/medial thigh
  - **Sacral plexus** (骶丛) L4-S4: posterior thigh, leg, foot
- Thoracic nerves are **intercostal nerves** (肋间神经)

Reflex Arcs

- **Reflex arc** (反射弧): neural pathway for a reflex
- Components:
  - **Sensory receptor** (感觉感受器)
  - **Sensory neuron** (感觉神经元)
  - **Integration center** (整合中枢)
  - **Motor neuron** (运动神经元)
  - **Effector** (效应器)
- Rapid, automatic, involuntary responses

Types of Reflexes

- **Monosynaptic reflex** (单突触反射): one synapse
  - Example: **knee-jerk reflex** (膝反射)
  - Simplest and fastest
- **Polysynaptic reflex** (多突触反射): involves interneurons
  - Example: **withdrawal reflex** (撤退反射)
  - More complex, allows integration

Activation of Motor Neurons

- **Somatic motor neurons** (躯体运动神经元) activated by:
  - Direct sensory neuron stimulation (reflex arc)
  - Spinal **association neurons/ interneurons** (联络神经元/中间神经元)
  - **Upper motor neurons** (上运动神经元) from brain motor areas
- Allows voluntary control to modify or override reflexes
- Integration provides flexibility in motor responses

Clinical Significance of Reflexes

- Reflexes diagnostically important for nervous system disorders:
  - **Hyperreflexia** (反射亢进): exaggerated reflexes, upper motor neuron damage
  - **Hyporeflexia** (反射减弱): diminished reflexes, lower motor neuron damage
  - **Babinski reflex** (巴宾斯基反射): normal in infants, abnormal in adults (pyramidal tract damage)
- Different reflex tests assess specific spinal levels
- Helps localize nervous system lesions

Summary

- The CNS is organized hierarchically from spinal cord through brain stem to cerebrum
- Each level has specialized functions
- Integration across levels enables complex behaviors
- Understanding structure-function relationships is essential for diagnosing and treating neurological disorders
- The brain's plasticity allows learning, memory, and recovery from injury